Improved Documentation for Node and Element Mapping Process

1. Objective and Context

The objective of this process is to establish a systematic and consistent identification (ID) and mapping system for structural nodes and elements in the wing model of an aircraft. This methodology ensures clarity, traceability, and compatibility across different stages of the finite element model (FEM) construction and analysis. Proper node and element mapping simplifies the representation of the wing structure and facilitates accurate simulation, analysis, and validation of results.

2. Node Organization and Naming Convention

We have a series of 2D coordinate datasets representing different sets of structural nodes. These nodes are organized into groups, each corresponding to specific structural components or regions of the wing. The naming conventions reflect their physical location and structural role:

- **nodos_larguerillos**: Nodes corresponding to stringers (larguerillos) distributed across the wing span.
- **nodos_posterior_ala**: Nodes corresponding to the rear spar of the wing, ordered from root to tip.
- **nodos_anterior_ala**: Nodes corresponding to the front spar of the wing, ordered similarly.
- **barras_ala_larguero_anterior**: Structural bars connecting nodes along the front spar.
- **nodos_larguerillos_fuselaje**: Nodes connecting the stringers to the fuselage.

Each of these node sets is initially defined in a two-dimensional coordinate system (Nx2 matrix), where:

- Column 1: X-coordinate
- Column 2: Y-coordinate

3. Step 1: Assigning Global Node IDs

Using the assignNodeIDs function:

- Each local node group (e.g., rear spar, front spar, stringers) is assigned a **global node ID** systematically.
- Local node IDs within each group are sequential and specific to that group.

- Global node IDs ensure each node is uniquely identifiable across the entire model.

Output from this step:

- **GlobalNodes Matrix:** A combined Nx3 matrix where each row represents:
    - Column 1: Global Node ID
    - Column 2: X-coordinate
    - Column 3: Y-coordinate
- **LocalToGlobalMap Cell Array:** A mapping matrix for each node group showing the correspondence between local and global IDs.

This step standardizes node identification across all node groups and prevents overlaps or inconsistencies when assembling the global model.

4. Step 2: Mapping Local Elements to Global IDs

Using the mapLocalToGlobalElements function:

- Structural elements defined locally (e.g., bars between nodes, connections) are updated to reference **global node IDs** instead of local ones.
- Each local element set is mapped to global IDs using the **LocalToGlobalMap** created in the previous step.
- A dictionary structure (containers.Map) ensures efficient and error-free mapping of node IDs.

Output from this step:

- **GlobalElements Cell Array:** A set of matrices representing structural elements, where each node reference uses a **global ID**.

5. Importance of the Node and Element Mapping System

- **Consistency:** Avoids duplicate or conflicting node identifiers across different structural sets.
- **Traceability:** Facilitates identification and debugging of specific nodes and elements during analysis.
- **Scalability:** Allows seamless addition or modification of structural components without disrupting the global ID system.
- **Compatibility:** Ensures smooth integration with finite element analysis (FEA) tools such as Nastran/Patran.

6. Next Steps

After successfully mapping nodes and elements:

- Define boundary conditions and loads on global nodes.
- Validate the node-element connectivity matrix.
- Proceed to finite element meshing and analysis.

This documentation outlines the foundation for structured FEM modeling of the aircraft wing, ensuring clarity, precision, and robust organization of nodes and elements.